

Q1. Bootstrapping and Confidence Intervals – Point Estimate

```
# Delivery-time data
```

```
data <- c(31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36)
```

```
# Sample size
```

```
n <- length(data)
```

```
# Point estimate: Sample mean
```

```
observed_mean <- mean(data)
```

```
# Print sample mean
```

```
cat("Sample Mean =", observed_mean, "minutes\n")
```

```
# Set seed for reproducibility
```

```
set.seed(123)
```

```
# Number of bootstrap resamples
```

```
B <- 10000
```

```
# Generate bootstrap sample means
```

```
bootstrap_means <- replicate(B, {
```

```
  bootstrap_sample <- sample(data, size = n, replace = TRUE)
```

```
  mean(bootstrap_sample)
```

```
})
```

```

# 95% Bootstrap Confidence Interval

CI <- quantile(bootstrap_means, probs = c(0.025, 0.975))

# Display confidence interval

cat("95% Bootstrap Confidence Interval:\n")

print(CI)

# -----

# Histogram of bootstrap distribution

# -----

hist(bootstrap_means,
     breaks = 40,
     probability = TRUE,
     main = "Bootstrap Distribution of Sample Mean",
     xlab = "Bootstrap Sample Mean (minutes)",
     ylab = "Density")

# Add density curve

lines(density(bootstrap_means),
      lwd = 2)

# Mark observed sample mean

abline(v = observed_mean,
       lwd = 2,

```

```
lty = 2)
```

```
# Mark confidence interval limits
```

```
abline(v = CI[1],
```

```
lwd = 2,
```

```
lty = 3)
```

```
abline(v = CI[2],
```

```
lwd = 2,
```

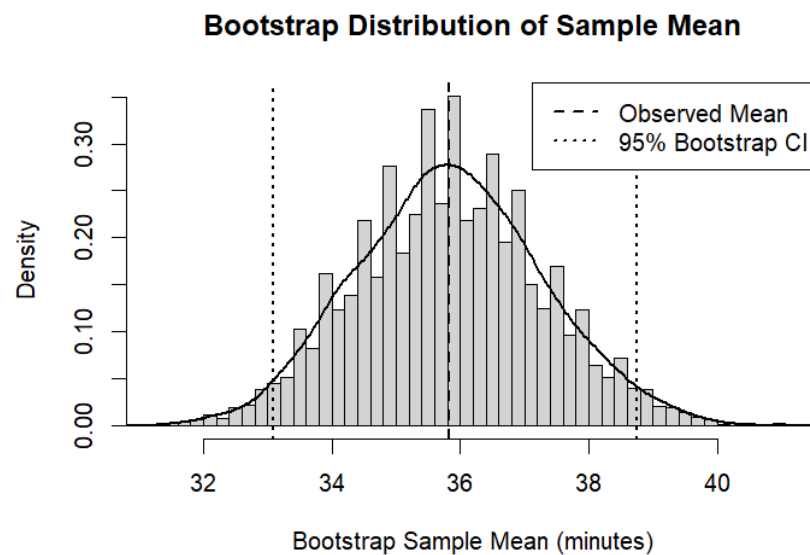
```
lty = 3)
```

```
# Add legend
```

```
legend("topright",
```

```
legend = c("Observed Mean",
```

```
"95% Bootstrap CI"),
```



Q2. Uncertainty in Trend Estimation

Objective

Fit a linear regression model between **Advertising Expenditure** and **Sales**, plot the fitted regression line with a **95% confidence band**, and explain how uncertainty changes across the range.

R Code

```
# Install packages if required
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Data
```

```
Advertising <- c(10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40)
```

```
Sales <- c(48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84)
```

```
# Create data frame
```

```
data <- data.frame(Advertising, Sales)
```

```
# Fit linear regression model
```

```
model <- lm(Sales ~ Advertising, data = data)
```

```
# Display regression results
```

```
summary(model)
```

```
# Regression coefficients
```

```
coef(model)
```

```
# Create confidence and prediction intervals
```

```
new_data <- data.frame(  
  Advertising = seq(min(Advertising),  
                    max(Advertising),  
                    length.out = 100)  
)
```

```
# 95% confidence interval for mean sales
```

```
confidence <- predict(model,  
  newdata = new_data,  
  interval = "confidence",  
  level = 0.95)
```

```
# 95% prediction interval for individual sales
```

```
prediction <- predict(model,  
  newdata = new_data,  
  interval = "prediction",  
  level = 0.95)
```

```
# Add confidence interval values
```

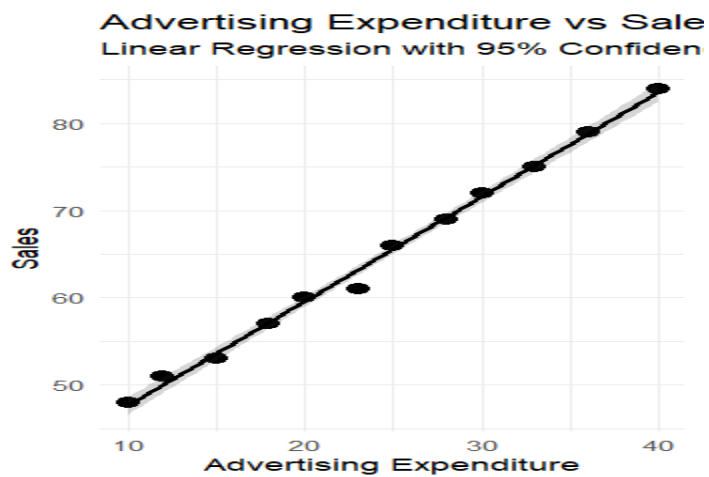
```
new_data$fit <- confidence[, "fit"]  
new_data$lwr <- confidence[, "lwr"]  
new_data$upr <- confidence[, "upr"]
```

```
# Plot observations, regression line and 95% confidence band
```

```

ggplot(data, aes(x = Advertising, y = Sales)) +
  geom_point(size = 3) +
  geom_ribbon(data = new_data,
            aes(x = Advertising,
                ymin = lwr,
                ymax = upr),
            alpha = 0.2,
            inherit.aes = FALSE) +
  geom_line(data = new_data,
            aes(x = Advertising, y = fit),
            linewidth = 1) +
  labs(
    title = "Advertising Expenditure vs Sales",
    subtitle = "Linear Regression with 95% Confidence Band",
    x = "Advertising Expenditure",
    y = "Sales"
  ) +
  theme_minimal()

```



Regression Equation

The fitted model will be approximately:

$$\widehat{Sales} = 34.9 + 1.22(Advertising)$$

This indicates that for every **one-unit increase in advertising expenditure**, predicted sales increase by approximately **1.22 units**.

Interpretation of Uncertainty

The **95% confidence band is narrowest near the center of the observed advertising range** because there is more information around the mean advertising expenditure.

The uncertainty increases toward the lower and upper extremes of the range. Therefore:

- Predictions near the center are more precise.
- Predictions near the boundaries have greater uncertainty.
- The confidence band represents uncertainty about the **mean sales trend**.
- A prediction interval for an individual future observation is wider because it includes both uncertainty in the estimated trend and individual variation.

Conclusion: There is a clear positive linear relationship between advertising expenditure and sales, but the estimated trend becomes less certain toward the edges of the observed advertising range.

Q3. Overplotting, Jittering and Transparency

Objective

Create:

1. A normal scatterplot.
2. A scatterplot using transparency and jittering.
3. Compare the two visualizations.

R Code

```
library(ggplot2)
```

```
# Data

Hours <- c(2,3,3,4,4,4,5,5,5,5,6,6,6,6,
          7,7,7,7,7,8,8,8,8,8,9,9,9,10,10)

Scores <- c(45,52,52,58,60,60,60,64,66,66,66,
           70,72,72,72,72,76,78,78,80,80,
           84,84,85,85,85,88,90,90,94,94)
```

```
data <- data.frame(Hours, Scores)
```

```
# -----
```

```
# Plot 1: Original scatterplot
```

```
# -----
```

```
ggplot(data, aes(x = Hours, y = Scores)) +
  geom_point(size = 3) +
  labs(
    title = "Original Scatterplot",
    x = "Hours Studied",
    y = "Examination Score"
  ) +
  theme_minimal()
```

```
# -----
```

```
# Plot 2: Transparency + Jittering
```

```
# -----
```



```

ggplot(data, aes(x = Hours, y = Scores)) +
  geom_jitter(
    width = 0.15,
    height = 0.8,
    alpha = 0.5,
    size = 3
  ) +
  labs(
    title = "Scatterplot with Jittering and Transparency",
    x = "Hours Studied",
    y = "Examination Score"
  ) +
  theme_minimal()

```



Interpretation

In the original scatterplot, several students have identical **Hours–Score combinations**. Therefore, points are drawn directly on top of one another.

This causes **overplotting**.

For example, multiple students with:

- 3 hours and 52 marks
- 5 hours and 66 marks
- 6 hours and 72 marks
- 8 hours and 85 marks

may appear as only a single point.

Consequently, the original plot can **underestimate the number of observations at common values**.

Jittering slightly moves overlapping points so that individual observations become visible. Transparency also allows darker regions to indicate where multiple points overlap.

Comparison

Original plot	Jitter + transparency
Overlapping points are hidden	Individual observations become visible
Frequency is difficult to judge	Concentration is easier to identify
Dense regions may appear sparse	Dense regions become visually obvious
Exact values are preserved	Small artificial displacement is introduced

Conclusion: The jittered and transparent plot communicates the data more effectively because it reveals both the overall relationship and the concentration of repeated observations.

Q4. 2D Histogram and Density Estimation

Objective

Visualize the relationship between **height and weight** using:

1. A 2D histogram/binning plot.
2. A 2D density plot.

Then identify regions with high observation density.

R Code

```
library(ggplot2)

# Data

Height <- c(150,152,155,158,160,161,163,165,166,168,
            170,171,173,175,176,178,180,182,184,186)

Weight <- c(48,50,53,55,57,59,60,62,64,65,
            68,69,71,73,74,77,80,82,85,88)

data <- data.frame(Height, Weight)

# -----

# Plot 1: 2D Histogram

# -----

ggplot(data, aes(x = Height, y = Weight)) +
  geom_bin2d(bins = 6) +
  labs(
    title = "2D Histogram of Height and Weight",
    x = "Height (cm)",
    y = "Weight (kg)"
  ) +
```

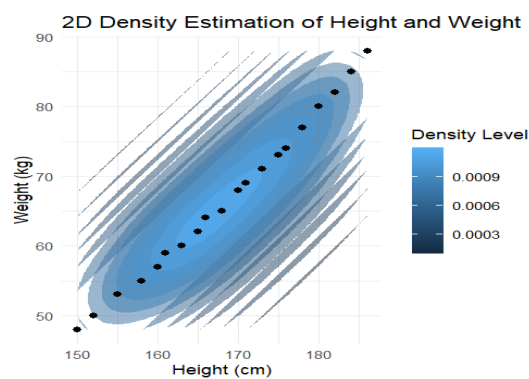
```
theme_minimal()
```

```
# -----
```

```
# Plot 2: 2D Density
```

```
# -----
```

```
ggplot(data, aes(x = Height, y = Weight)) +  
  stat_density_2d(  
    aes(fill = after_stat(level)),  
    geom = "polygon",  
    alpha = 0.5  
  ) +  
  geom_point(size = 2) +  
  labs(  
    title = "2D Density Estimation of Height and Weight",  
    x = "Height (cm)",  
    y = "Weight (kg)",  
    fill = "Density Level"  
  ) +  
  theme_minimal()
```



Q5. Color Scales, Accessibility and Misleading Visualization

R Code

```
library(ggplot2)

# Data

Category <- c("A", "B", "C", "D", "E", "F", "G")
Satisfaction <- c(42, 48, 55, 61, 67, 74, 82)

data <- data.frame(Category, Satisfaction)

# -----

# Plot 1: Potentially misleading
# Diverging/nonmonotonic color scheme
# -----

ggplot(data,
  aes(x = Category,
      y = Satisfaction,
      fill = Satisfaction)) +
  geom_col() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
```

```

midpoint = 62
) +
labs(
  title = "Satisfaction Scores - Diverging Color Scheme",
  x = "Category",
  y = "Satisfaction Score",
  fill = "Satisfaction"
) +
theme_minimal()

```

Improved Sequential Visualization

A color-vision-deficiency-friendly sequential scale can be created using **viridis**.

```
# install.packages("viridis") # Run once if needed
```

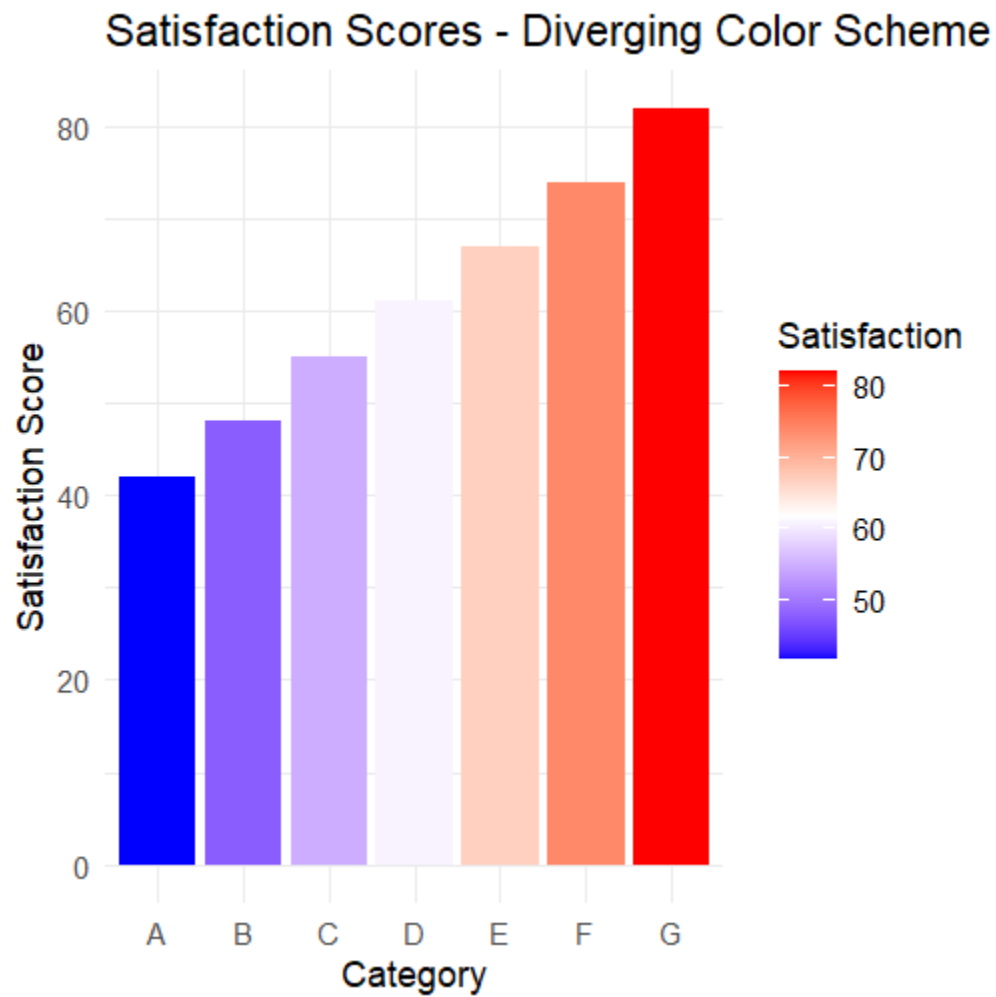
```

library(viridis)

ggplot(data,
  aes(x = Category,
      y = Satisfaction,
      fill = Satisfaction)) +
  geom_col() +
  scale_fill_viridis(
    option = "viridis",
    direction = 1
  ) +
  labs(
    title = "Satisfaction Scores - Sequential Color Scale",

```

```
x = "Category",
y = "Satisfaction Score",
fill = "Satisfaction"
) +
theme_minimal()
```



Interpretation

The satisfaction scores increase steadily:

Category	Satisfaction
----------	--------------

A	42
---	----

Category Satisfaction

3	18
3	55
3	51
3	57
3	74
3	82

The first visualization uses a **diverging color scale** where lower values can appear blue, middle values white, and higher values red.

This is potentially misleading because a diverging color scale implies a meaningful **central reference point or two-sided interpretation**. Here, satisfaction is an ordered variable with a natural direction from low to high. There is no substantive reason for the midpoint to be treated as a special boundary.

The second visualization uses the **viridis sequential scale**, where color changes progressively with the numerical value.

Why the Second Design Is Better

The sequential scale is more appropriate because:

- Satisfaction is an ordered numerical variable.
- Color intensity progresses consistently with the score.
- There is no artificial visual emphasis around an arbitrary midpoint.
- The scale is designed to remain relatively distinguishable for people with common forms of color-vision deficiency.
- It makes the low-to-high ordering easier to interpret.

Conclusion: The main visualization pitfall in the first plot is the use of a **diverging/nonmonotonic color scheme for inherently ordered data without a meaningful midpoint**. A sequential scale provides a clearer and more accessible representation.
